Release Notes

for S32K342_SBAF_HSEB_RTM_0.15.0

Rev. 1.0 — 1 September 2023

Release notes

COMPANY CONFIDENTIAL

1 Getting Started

1.1 Package content

This package contains the NXP S32K342 SBAF 0.15.0:

- FULL_MEM SBAF: encrypted binary
- The license.txt file and the `uninstall.exe` utility for removing the SBAF binaries
- SBAF_S32K342_0_0_15_0_ReleaseNotes.pdf - this file

2 Release Details

This is the HSE Secure-BAF (SBAF) 0.15.0 RTM release for S32K342 for FULL_MEM configuration. The SBAF release of AB_SWAP configuration is part of the HSE firmware AB_SWAP pink image. Both releases can be used for production.

It is strongly recommended that user should update the SBAF in their existing samples if the features provided in this release are important to them.

The SBAF can be updated only in FULL_MEM configuration. In case of AB_SWAP, the SBAF gets updated as part of the HSE firmware update.

This release was developed and tested using:

- Chip : P32K312NGVPBS
- Mini-Module : XS32K3X2CVB-Q172

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP:

- S32K342
- S32K322
- S32K341

2.2 Version Number Details

Version number is of 8 bytes size and can be read at HSE GPR address: 0x4039C020 - - 0x4039C027

FULL_MEM:

00 0D 00 00 00 0F 00 06

AB_SWAP:

01 0D 00 00 00 0F 00 05

3 Secure-BAF Update

Secure-BAF (SBAF) can be updated via `HSE_SRV_ID_SBAF_UPDATE` and service structure "`hseSbafUpdateSrv_t`" provided by HSE FW in one shot or in streaming mode.

The new encrypted SBAF image can be programmed at any address in the application memory before calling the SBAF update service.

The encrypted SBAF image is authenticated and decrypted, then stored in the HSE code flash memory.

On the next reset, the new SBAF boots the HSE Firmware.

Note:

- *NXP recommends updating SBAF in one shot mode.*
- *SBAF update must be performed in a stable environment because any failure in the update process causes the bricking of device.*
- *SBAF update service is only applicable for FULL_MEM configuration.*

Please refer to the **HSE Firmware Reference Manual** for more details.

4 Changes in RTM 0.15.0

Features Added

- Error-Correcting Code (ECC) handling feature:This feature will prevent the bricking of the device in case reset happens during secure flash area programming.
- Added the feature to make the desired area of application code flash area as OTP. This feature can be used only if corresponding service is provided in HSE Firmware. Currently, this feature is *not supported* by the standard HSE Firmware.
- Added the support of enabling the Firmware feature flag in Boot configuration word (Bit number 9) in addition to UTEST location 0x1B000000. The same bit can enable the firmware installation mode via MU without the need to write the marker value 0xA5 in DCM register. For more details, refer to the **HSE Firmware reference manual**.
- Added the feature in SBAF to support the firmware installation and update in backup feature enable and disable configuration via a bit 10 in Boot configuration word in IVT. This feature is applicable only for FULL_MEM configuration as in case of AB_SWAP, SBAF and HSE firmware are combined as single image. The backup disable feature in currently *not supported* in standard HSE firmware.
- Added the feature to disable the option to put application core in recovery mode in case of 8 functional and 8 destructive resets via bit number 8 in Boot configuration word in IVT. This feature is applicable only for FULL_MEM configuration as backup. In case of AB_SWAP, backup is always taken in passive area not in data flash.

5 Legal information

5.1 Definitions

Draft — A draft status on a document indicates that the content is still under internal review and subject to formal approval, which may result in modifications or additions. NXP Semiconductors does not give any representations or warranties as to the accuracy or completeness of information included in a draft version of a document and shall have no liability for the consequences of use of such information.

5.2 Disclaimers

Limited warranty and liability — Information in this document is believed to be accurate and reliable. However, NXP Semiconductors does not give any representations or warranties, expressed or implied, as to the accuracy or completeness of such information and shall have no liability for the consequences of use of such information. NXP Semiconductors takes no responsibility for the content in this document if provided by an information source outside of NXP Semiconductors.

In no event shall NXP Semiconductors be liable for any indirect, incidental, punitive, special or consequential damages (including - without limitation - lost profits, lost savings, business interruption, costs related to the removal or replacement of any products or rework charges) whether or not such damages are based on tort (including negligence), warranty, breach of contract or any other legal theory.

Notwithstanding any damages that customer might incur for any reason whatsoever, NXP Semiconductors' aggregate and cumulative liability towards customer for the products described herein shall be limited in accordance with the Terms and conditions of commercial sale of NXP Semiconductors.

Right to make changes — NXP Semiconductors reserves the right to make changes to information published in this document, including without limitation specifications and product descriptions, at any time and without notice. This document supersedes and replaces all information supplied prior to the publication hereof.

Suitability for use — NXP Semiconductors products are not designed, authorized or warranted to be suitable for use in life support, life-critical or safety-critical systems or equipment, nor in applications where failure or malfunction of an NXP Semiconductors product can reasonably be expected to result in personal injury, death or severe property or environmental damage. NXP Semiconductors and its suppliers accept no liability for inclusion and/or use of NXP Semiconductors products in such equipment or applications and therefore such inclusion and/or use is at the customer's own risk.

Applications — Applications that are described herein for any of these products are for illustrative purposes only. NXP Semiconductors makes no representation or warranty that such applications will be suitable for the specified use without further testing or modification.

Customers are responsible for the design and operation of their applications and products using NXP Semiconductors products, and NXP Semiconductors accepts no liability for any assistance with applications or customer product design. It is customer's sole responsibility to determine whether the NXP Semiconductors product is suitable and fit for the customer's applications and products planned, as well as for the planned application and use of customer's third party customer(s). Customers should provide appropriate design and operating safeguards to minimize the risks associated with their applications and products.

NXP Semiconductors does not accept any liability related to any default, damage, costs or problem which is based on any weakness or default in the customer's applications or products, or the application or use by customer's third party customer(s). Customer is responsible for doing all necessary testing for the customer's applications and products using NXP Semiconductors products in order to avoid a default of the applications and the products or of the application or use by customer's third party customer(s). NXP does not accept any liability in this respect.

Terms and conditions of commercial sale — NXP Semiconductors products are sold subject to the general terms and conditions of commercial sale, as published at http://www.nxp.com/profile/terms, unless otherwise agreed in a valid written individual agreement. In case an individual agreement is concluded only the terms and conditions of the respective agreement shall apply. NXP Semiconductors hereby expressly objects to applying the customer's general terms and conditions with regard to the purchase of NXP Semiconductors products by customer.

Export control — This document as well as the item(s) described herein may be subject to export control regulations. Export might require a prior authorization from competent authorities.

Suitability for use in non-automotive qualified products — Unless this document expressly states that this specific NXP Semiconductors product is automotive qualified, the product is not suitable for automotive use. It is neither qualified nor tested in accordance with automotive testing or application requirements. NXP Semiconductors accepts no liability for inclusion and/or use of non-automotive qualified products in automotive equipment or applications.

In the event that customer uses the product for design-in and use in automotive applications to automotive specifications and standards, customer (a) shall use the product without NXP Semiconductors' warranty of the product for such automotive applications, use and specifications, and (b) whenever customer uses the product for automotive applications beyond NXP Semiconductors' specifications such use shall be solely at customer's own risk, and (c) customer fully indemnifies NXP Semiconductors for any liability, damages or failed product claims resulting from customer design and use of the product for automotive applications beyond NXP Semiconductors' standard warranty and NXP Semiconductors' product specifications.

Translations — A non-English (translated) version of a document, including the legal information in that document, is for reference only. The English version shall prevail in case of any discrepancy between the translated and English versions.

Security — Customer understands that all NXP products may be subject to unidentified vulnerabilities or may support established security standards or specifications with known limitations. Customer is responsible for the design and operation of its applications and products throughout their lifecycles to reduce the effect of these vulnerabilities on customer's applications and products. Customer's responsibility also extends to other open and/or proprietary technologies supported by NXP products for use in customer's applications. NXP accepts no liability for any vulnerability. Customer should regularly check security updates from NXP and follow up appropriately.

Customer shall select products with security features that best meet rules, regulations, and standards of the intended application and make the ultimate design decisions regarding its products and is solely responsible for compliance with all legal, regulatory, and security related requirements concerning its products, regardless of any information or support that may be provided by NXP.

NXP has a Product Security Incident Response Team (PSIRT) (reachable at PSIRT@nxp.com) that manages the investigation, reporting, and solution release to security vulnerabilities of NXP products.

5.3 Trademarks

Notice: All referenced brands, product names, service names, and trademarks are the property of their respective owners.

NXP — wordmark and logo are trademarks of NXP B.V.

Release Notes

for S32K342_SBAF_HSEB_RTM_0.15.0

Contents

For more information, please visit: http://www.nxp.com

Date of release: 1 September 2023
Document identifier: BLN_S32K342_SBAF_HSEB_RTM_0.15.0